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Applications of Multiagent Systems in Game Theory

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Abstract

This research paper is presented as an introduction for the applications of Multiagent Systems (MAS) in the field of Game Theory (GT), with a focus on how autonomous agents can model strategic decision making in complex environments. MAS and GT are strongly connected because both study the interaction between multiple independent entities that must cooperate, compete, negotiate or coordinate to achieve their objectives.

The paper begins by introducing the fundamental concepts of MAS and GT, together with the main game structures such as cooperative and non-cooperative games, zero-sum and non-zero-sum games and sequential and simultaneous games.

To better understand these concepts, two representative case studies are analysed: the Minimum Effort Game as a coordination problem and the Vickrey Auction as a competitive decision making model. Both simulations are implemented using multi-agent approaches and are compared with existing literature results. The study shows how MAS can provide efficient solutions for strategic interactions while also helping researchers simulate, analyse and validate game theoretic behaviour in real-world scenarios.

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Keywords: (MAS) Multiagent Systems; (GT) Game Theory; (AI) Artificial Intelligence; (DL) Deep Learning (ML) Machine Learning; Coordination Games (GT); Minimum Effort Game (MEG); Vickrey Auction (VA)

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1. Introduction

1.1. Study Motivation

This research paper is driven by my interest in Game Theory (GT) and its applications in **mathematics**, **economics** and **computer science**. During my studies, I have come to understand how decision-making strategies can be expressed using mathematical models and applied to real-world situations involving multiple interacting agents. GT helps explain how agents compete, cooperate and reach stable outcomes, which makes it useful in areas like markets, distributed systems and algorithm design. Based on this, this paper aims to explore how Multiagent Systems (MAS) can be used within GT, focusing on practical applications where autonomous agents use strategic reasoning to solve real-world problems.

1.2. Core Concepts

The main concept of the present research paper regards MAS. There are various written papers defining what a MAS, starting with a really valid reference on the book "*An introduction to multiagent systems*" of **Wooldridge Michael** [21]. The book states, by citing, "an agent is a computer system that is capable of independent action on behalf of its user or owner. ... a multiagent system is one that consists of a number of agents, which interact with one another, typically by exchanging messages through some computer network infrastructure".

This concept is consolidated over various studies, in fact, another valid reference is "*Multiagent systems: a modern approach to distributed artificial intelligence*" of "Weiss Gerhard" [20], defining MAS as "systems in which several interacting, intelligent agents pursue some set of goals or perform some set of tasks". This study expands MAS within Artificial Intelligence Artificial Intelligence (AI) field.

The second core concept of this research paper is GT, the study of mathematical models that describe how rational decision-makers interact strategically, as described in the book "**Game Theory**" of "*Myerson, R.B.*" [10]. While GT originated as a way to analyze economic behaviors, it has grown into a highly versatile tool used across the social sciences and economy [16]. Today, it can also serve in computer science field for understanding MAS, where multiple autonomous entities must negotiate, compete or cooperate to achieve their objectives.

Historically, GT began by looking at simple two person "*zero-sum*" games scenarios where one participant's gain is exactly tied to the other participant's loss. However, as the science evolved through the 1950s, it expanded to include "*non-zero-sum*" games, becoming a broad science of decision making for humans, animals and machines.

The birth of modern GT is credited to **John von Neumann**, who first proved the existence of mixed strategy equilibria. His groundbreaking mathematical proofs were further developed in the seminal 1944 book, "**Theory of Games and Economic Behavior**" [19], co-authored with Oskar Morgenstern. This work pushed the field beyond strict two-player conflicts and introduced the study of cooperative games involving multiple players. Building upon these foundational theories, this paper will explore how modern MAS apply game theoretic concepts to navigate complex and interactive environments.

2. Literature Review

2.1. Game Theory Structures and Interaction Models

Within the literature, GT is systematically categorized into several core interaction formats that are widely used to model multiagent environments. There are various distinctions between GT structures, as following:

- **Cooperative vs. Non-cooperative games:** A cooperative game is a game where players can form a binding agreement, which is a strict contract to enforce an action. In these games, subsets of players form a coalition, or a group that acts together, to gain a better payoff than they could alone. The goal of cooperative game theory is to determine the payoff a group receives when they work together to improve their collective welfare, as exposed in article "**Cooperative Game Theory**" of "*Chandrasekaran R.*" [2].
  In contrast, a non-cooperative game is a game where decision-makers cannot make binding agreements, as exposed in book "**Non-Cooperative Game Theory**" by "*Takako Fujiwara-Greve*" [6]. Each player acts independently without collaborating or communicating with others. The goal of non-cooperative game theory is to find out what happens when individuals make independent, strategic decisions. Non-cooperative games focus on independent decision-making and equilibrium outcomes. An equilibrium is a stable result where players have made their best choices. A common outcome is the Nash equilibrium, which is a state where no player can get a better payoff by changing their strategy alone;
- **Symmetric vs. Asymmetric games:** In symmetric games, players have identical roles and receive the same payoffs when choosing the same strategies. A symmetric game requires all players to share the exact same options for strategies. Additionally, the rewards must be permutation-invariant, which means that swapping players' choices swaps their rewards exactly. These concepts are detailed in the article "**Equilibria in symmetric games: Theory and applications**" by "*Andreas Hefti*" [8]. In contrast, asymmetric games involve differences in strategy sets or payoff structures between players. An asymmetric game occurs when the rules for rewards differ across players due to different parameters. Even in a symmetric game, identical players might choose different actions, which creates an asymmetric equilibrium. An equilibrium is a stable state where no player has an incentive to change their strategy;

		Player 2	
		E	F
Player 1	E	1, 2	0, 0
	F	0, 0	1, 2

Table 1. Sample asymmetric game payoff schema

- **Zero-sum vs. Non-zero-sum games:** A zero-sum game is a strictly competitive situation. In this game, one player's gain is exactly equal to another player's loss, as exposed in book "**Game Theory**" by "*Guillermo Owen*" [12]. The total amount of reward does not change. Mathematically, the sum of all payoffs is zero: $\sum_{i=1}^{n} u_i = 0$. Here, n is the total number of players, and u_i is the payoff (the reward or outcome) for player i. A simple example is a coin-flipping bet. If you win a dollar, your friend loses exactly one dollar. In contrast, a non-zero-sum game allows for results where players can win together or lose together (sample: Prisoner's Dilemma);
- **Simultaneous vs. Sequential games:** [3] Simultaneous games involve players making decisions at the same time without knowledge of others' actions, while sequential games unfold over time, where earlier actions influence later decisions.
- **Simultaneous vs. Sequential games:** Simultaneous games involve players making decisions at the same time without knowing what others will choose. On the other part, a sequential game unfold over time, where earlier actions influence later decisions. As exposed in "**Design Theory and Methods Using CAD**" by "*Chang Kuang-*

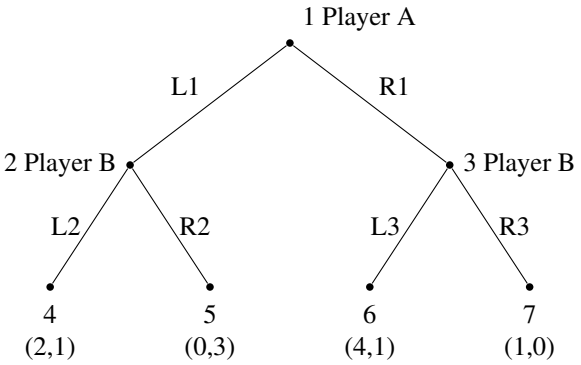
		Player 2	
		A	B
Player 1	A	-1, 1	3, -3
	B	0, 0	-2, 2

Table 2. Sample zero-sum game payoff schema

Hua" [3], it is possible to map out a sequential game using a **game tree** (also called the **extensive form**). A game tree has three main parts:

- **Nodes (or vertices):** Points where a specific player must make a decision. Each node is owned by exactly one player;
- **Arrows:** Lines coming out of a node that represent the possible actions a player can choose;
- **Terminal nodes:** The bottom points of the tree that show the final rewards (payoffs) for everyone.

Sequential games often assume **perfect information**. This means when it is a player's turn to move, they already know all the decisions that the previous players have made.



Sample zero-sum game payoff schema

2.2. *Game Theory Applications in Multiagent Systems*

There are some papers and studies researching and exposing GT applications in MAS. A good reference in this case is the article "**Game Theory and Decision Theory in Multi-Agent Systems**" by "*Parsons Simon and Wooldridge Michael*". They express that GT and MAS **integrate naturally**, so by combining the two fields, researchers can program agents to negotiate, coordinate and make optimal decisions in complex environments [17].

Based on Parsons and Wooldridge research, there are several major applications where these two fields work together:

1. **Electronic Commerce (E-Commerce)**: E-commerce is the buying and selling of goods over the internet. In this field, agents use game theory to act as automated buyers and sellers.

   - **Shopbots:** A shopbot is an agent that works for a human buyer. It searches the internet to compare prices across many websites. It uses game-theoretic strategies to find the best quality product for the lowest possible cost;
   - **Pricebots:** A pricebot is an agent that works for a seller. It automatically adjusts the price of a product based on market demand. Pricebots use learning algorithms to maximize the seller's profit while avoiding a "price war", a negative situation where competing sellers repeatedly lower their prices to beat each other.

2. **Automated Negotiation**: Automated negotiation is a process where computer programs reach an agreement without human help. Because agents are self-interested, they must compromise to achieve their goals. Game theory provides the exact mathematical rules for this process. It allows agents to calculate their preferences and choose a strategy that guarantees the best possible outcome during a negotiation.

3. **Bounded Rationality and Satisficing**: Classical game theory assumes that players have unlimited time to calculate the perfect move. In computer science, this is impossible. Computers have limits on memory and processing speed.

   - **Bounded Rationality:** This is the concept that agents have limited computing resources and cannot always find the perfect mathematical solution;
   - **Satisficing:** To solve this problem, agents use satisficing. This means the agent stops searching for the perfect solution as soon as it finds a solution that is "good enough."

   To do this, agents calculate a special score called "praxeic utility", an utility thet balances the value of a good decision against the high cost of the computing power needed to find it.

4. **Hyper-games for Uncertain Environments**: In many situations, an agent does not know exactly what its opponent is doing. To handle this uncertainty, agents use a framework called hyper-games. A hyper-game is a model where an agent builds a probability distribution over several different possible games. Instead of assuming it is playing one specific game, the agent hedges its bets. It calculates the most likely behaviours of its opponent and evaluates the best average move across all those possibilities.

2.3. Multiagent Systems Applications in Game Theory

The scope of this research paper is to explore and expose proper applications of Multiagent Systems within GT field, with a particular focus on their use in simulation scenarios and case study analysis. Existing research approaches this topic from a MAS perspective. Notably, the work "*Game Theoretical Applications for Multi-Agent Systems*" by "*Parag C. Pendharkar*" [13] provides a structured overview of both cooperative and non-cooperative (competitive) game applications within multiagent environments:

- **Cooperative Multi-agent Systems**
  - **Description:** A cooperative MAS is designed so that agents always work together;
  - **When to use:** Suitable when a game has a dominant strategy, meaning the best choice for an agent remains optimal regardless of other agents' decisions;
  - **Problem:** The use of dominant strategies by all agents can lead to a social dilemma, where individually rational decisions produce a poor outcome for all participants;
  - **Solution:** Designers enforce cooperation and introduce convex preferences, ensuring agents avoid extreme actions such as monopolizing or giving away all resources;
  - **Example:** In a wireless network, cell towers act as agents sharing communication channels. If they behave selfishly, each tower handles fewer calls (*Nash Equilibrium*). Through cooperation, they can significantly increase total system capacity.

- **Competitive Multi-agent Systems**
  - **Description:** A competitive MAS (non-cooperative) consists of self-interested agents that prioritize their own objectives;
  - **When to use:** Appropriate for complex environments where no clear dominant strategies exist;
  - **Example:** In a factory scenario, two processors select jobs from a shared queue. If both select the same job simultaneously, it is lost. Without coordination, inefficiencies arise, so competitive rules (e.g., turn-taking) are applied to maintain system stability.

2.4. Multiagent Game Theory Games

As previously expressed, this research paper is concentrated on how MAS can be involved into simulation scenarios for GT games. There are several games in the field, but for this particular paper, only games with more than 2 players are of interest, in order to properly explore MAS scenarios. A list of valid GT games:

Game	Players	Strategies per player	Sequential	Zero sum
Vickrey auction	N	Infinite	No	No
Coordination game	N	Variable	No	No
El Farol bar	N	2	No	No
Diner's dilemma	N	2	No	No
Cake cutting	N (usually 2)	Infinite	Yes	Yes
Gift-exchange game	N (usually 2)	Variable	Yes	No
Guess 2/3 of the average	N	Infinite	No	Yes/No
Minimum effort (weak-link) game	N	Infinite	No	No
Muddy Children Puzzle	N	2	Yes	No
Peace war game	N	Variable	Yes	No
Pirate game	N	Infinite	Yes	No
Platonia dilemma	N	2	No	No
Public goods	N	Infinite	No	No
Signaling game	N	Variable	Yes	No
Volunteer's dilemma	N	2	No	No

Table 3. Selected Game Theory games and their characteristics

3. Related Work

3.1. Research Ideas

Since MAS is better represented and involved when included within multiple actors (players, participants, agents), for this paper study case "**Minimum Effort Game**" (as a Cooperation game) and "**Vickrey Auction**" (as a Conflict Game).

3.2. Coordination Game

A coordination game is a class of strategic interactions in which players achieve higher payoffs when they select compatible or identical strategies. According to "*Cooper Russell*" in his book "**Coordination games**" [4], these games are characterized by **strategic complementarities**, meaning that the benefit of choosing a particular action increases when other players choose the same action. Coordination games typically exhibit multiple Nash equilibria, where each equilibrium corresponds to a different coordinated outcome.

Formally, a strategy profile is called a *Nash equilibrium* if no player can improve their expected payoff by unilaterally deviating from their chosen strategy, given that the strategies of all other players remain fixed. This condition can be interpreted as a fixed point of the best-response correspondence over the product space of all players' strategy sets, as exposed by "*John F. Nash Jr.*" in his paper "**Equilibrium points in n-person games**" [11].

The fundamental mechanism of a coordination game relies on mutual expectations. Each player's optimal decision depends on their beliefs about the actions of others. If players successfully align their expectations, they converge toward a stable equilibrium. However, in the absence of communication or external guidance, players may fail to coordinate and reach suboptimal outcomes. This makes equilibrium selection a central challenge in coordination games.

A simple example of a coordination game is presented in Table 4. In this two-player scenario, both players choose between two strategies: A and B. The highest payoffs are obtained when both players select the same strategy, illustrating the need for coordination.

	Player 2	
Player 1	A	B
A	(2,2)	(0,0)
B	(0,0)	(1,1)

Table 4. Example of a coordination game

In this example, there are two Nash equilibria: (A, A) and (B, B). The equilibrium (A, A) is Pareto-superior, as it yields higher payoffs for both players, but achieving it requires successful coordination.

Coordination games are highly relevant in the context of MAS, where multiple autonomous agents operate without centralized control. In such systems, agents must align their decisions to ensure efficient system-wide outcomes. Coordination games provide a formal framework for modeling problems such as protocol selection, resource allocation and consensus formation. By leveraging learning algorithms, communication mechanisms or predefined conventions, agents can converge toward optimal equilibria, improving both individual and collective performance.

3.3. Minimum Effort Game

As a practical case study, the **Minimum Effort Game** is selected as a representative example of a Coordination Game, as it supports the inclusion of multiple agents and a range of strategic choices.

The Minimum Effort Game, also referred to as the weakest link game, is a model in which the participant with the lowest level of performance determines the overall outcome for all players involved. A common real-world example is an airport ground crew, as exposed on article "**Efficient coordination in weakest-link games**" of "*Arno Riedl, Ingrid M.T. Rohde and Martin Strobel*" [14]. An aircraft can depart only after all teams complete their assigned tasks, meaning that the total completion time depends on the slowest team.

In this setting, each player selects an effort level. The reward received by each player increases according to the lowest effort level chosen within the group. At the same time, each player incurs a cost that reduces their reward based on their individual effort level. As a result, the optimal strategy for each player is to match the minimum effort chosen by others in the group.

This relationship can be expressed using a formal equation. Let N denote the set of players, and let e_i represent the effort level selected by player i. The payoff (or final reward) for player i is given by

$$\pi_i(s) = a \cdot \min_{j \in N}\{e_j\} - b \cdot e_i + c$$

where a represents the benefit derived from the minimum group effort, b represents the cost associated with the player's own effort, and c is a constant added to ensure that payoffs remain non-negative. Since the parameters satisfy $a > b > 0$, each player is motivated to align their effort with the lowest effort level in the group.

When all players choose the same effort level, the system reaches an equilibrium. An equilibrium is a stable state in which no player can improve their payoff by changing their strategy independently. This model is particularly relevant for MAS, where multiple autonomous agents interact within a shared environment. The Minimum Effort Game provides a useful framework for analysing how such agents coordinate their actions and maintain system efficiency, especially in situations where a single weak performance can negatively affect the outcome for all participants.

3.3.1. Simulation Framework and Experimental Setup

A multi-agent simulation is used to study coordination in the Minimum Effort Game. It is implemented using an asynchronous agent-based system built on the *SPADE* framework, which supports message-based communication between distributed agents.

The system contains $N = 20$ agents and a central controller executing $H = 100$ repeated rounds. Each agent represents an autonomous decision-maker assigned a fixed strategy from a predefined set, including heuristics, reinforcement learning, regret-based learning, and game-theoretic best-response methods.

In each round, agents independently select an effort level $e_i \in \{1, \ldots, M\}$ with $M = 7$. The controller collects all actions, computes the minimum effort, and evaluates payoffs using the same utility function as the theoretical model. Results are then broadcast back to all agents to synchronize state.

Each agent updates its internal state using observed outcomes and applies its strategy-specific rule to choose the next action. Agents also store historical data on past efforts, minimum group values and payoffs.

Agentic strategies used per simulation:

- **Core heuristics**
  - `always_low`: selects the minimum effort (safe baseline behavior)
  - `always_high`: selects the maximum effort (high-risk, high-reward behavior)
  - `random`: selects effort uniformly at random
  - `imitate_min`: copies the previous round minimum effort
  - `gradual_up`: increases effort incrementally over time
  - `gradual_down`: decreases effort incrementally over time
  - `win_stay`: repeats action after high payoff
  - `lose_shift`: changes action after low payoff
- **Memory and predictive**
  - `predict_min`: predicts next-round minimum effort from history
  - `trend_following`: follows directional trends in past group behavior
  - `memory_3_step`: bases decision on last three rounds only
- **Game-theoretic**
  - `best_response`: best response to last observed actions
  - `fictitious_play`: responds to empirical distribution of past actions
  - `risk_dominant`: selects lowest-risk action
  - `regret_minimization`: chooses action minimizing historical regret
  - `stochastic_best_response`: best response with random perturbation
- **Learning and social**
  - `bandit_UCB`: upper confidence bound exploration-exploitation policy
  - `q_learning_low`: reinforcement learning maximizing expected payoff
  - `consensus_seek`: aligns with most frequent group action
- **Meta strategy**
  - `strategy_switcher`: dynamically switches strategy based on performance threshold

3.3.2. Literature Results Analysis

In the article "**The Optimal Strategy in the Minimum Effort Game**" by "*Edward Cartwright*" [1], the author explains that the best strategy in the Minimum Effort Game is not always choosing the highest effort level. Instead, players should choose the effort level that gives the best expected payoff based on what they believe other players will choose.

The study shows that in the first round, players choose many different effort levels and this behaviour does not change much with different group sizes. However, after repeated rounds, players usually move toward lower effort levels, especially in larger groups where coordination is harder. The author concludes that high effort levels are difficult to maintain in large groups because players expect coordination failure and prefer safer low effort choices.

Source	n	Proportion Choosing Effort Level							$E^{l,k}$	k^*
		1	2	3	4	5	6	7	$\sigma_i(k)$	
VHBB [28]	2	0.29	0.04	0.04	0.07	0.21	0.00	0.36	0.5	5
CGvV [36]	3	0.03	0.11	0.25	0.19	0.14	0.08	0.19	0.71	3
CGvV [96]	4	0.23	0.08	0.16	0.16	0.14	0.07	0.17	0.79	1
VHBB [28]	14	0.04	0.04	0.04	0.25	0.25	0.11	0.29	0.95	2
VHBB [15]	15	0.00	0.00	0.00	0.07	0.53	0.00	0.40	0.95	4
VHBB [64]	16	0.02	0.06	0.06	0.16	0.30	0.11	0.30	0.95	2
Average [267]		0.13	0.07	0.11	0.16	0.22	0.07	0.25		

Fig. 1. Minimum effort game choices of players in the first round. VHBB stands for Van Huyck, Battalio and Beil (1990) [9] and CGvV stands for Cartwright, Gillet and van Vugt [5].

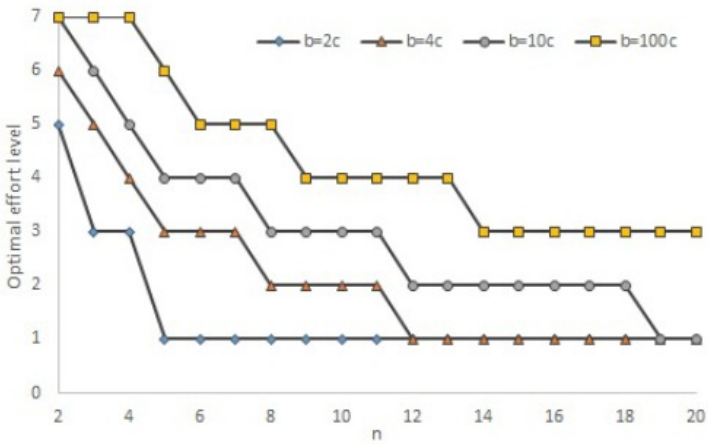


Fig. 2. Optimal effort level for "n" number of players

3.3.3. Results Analysis

The results show a consistent convergence pattern over the 100 simulation rounds. Effort levels progressively decline and stabilise at the minimum bound of the action space. This indicates strong coordination failure toward the lowest-effort equilibrium.

First, we analyse the Average Effort over time. At the beginning of the simulation, agents explore a wide range of effort levels, resulting in relatively high average effort. However, this quickly decreases as agents adapt their strategies. In later rounds, the system converges, and most agents consistently select the minimum effort level $e = 1$.

Second, the Effort Heat-map provides a finer view of individual behaviour. The heat-map encodes effort levels using colour intensity, where darker shades correspond to lower effort values. Over time, the distribution shifts toward darker regions, confirming a system-wide convergence toward minimal effort.

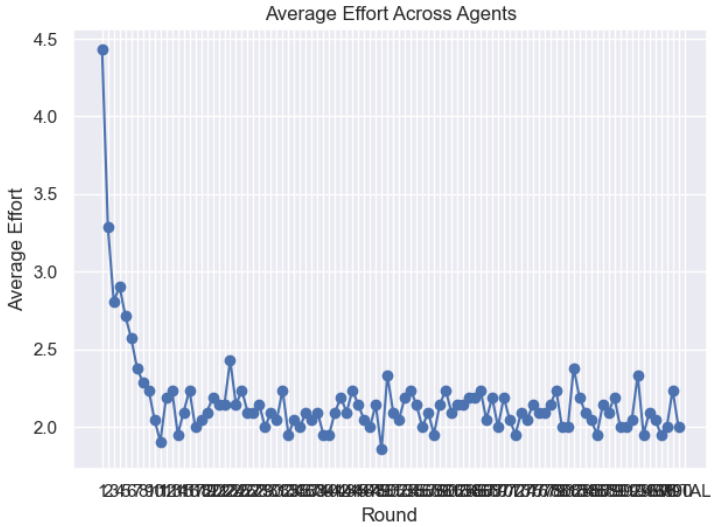


Fig. 3. Average effort evolution over 100 rounds

Fig. 4. Distribution of individual effort levels across rounds

Overall, the results indicate that high effort is not stable over time. Strategic adaptation drives agents to converge on the minimum effort level, which becomes the dominant equilibrium under the given payoff structure. The simulation shows that larger groups struggle to sustain coordination at higher effort levels. Due to the weak-link effect, agents learn that high effort is disadvantageous when others reduce their effort. As a result, they progressively lower their actions to avoid payoff losses, leading the system to converge quickly to the lowest effort state.

3.3.4. Comparison with Literature Review

The simulation results are consistent with the conclusions presented in the literature review, especially with the findings of Edward Cartwright [1]. Both the theoretical analysis and the simulation show that players tend to converge toward low-effort equilibria instead of maintaining socially efficient high-effort coordination.

3.4. Vickrey Auction

Vickrey Auction was first presented by "*William Vickrey*" in 1961 in his paper "**Counterspeculation, Auctions, and Competitive Sealed Tenders**" [18], also known as the second-price method. **A Vickrey Auction is a special type of sealed-bid auction**, a process where all participants submit their offers secretly. In this game, the rules state that the award is made to the highest bidder, but the price paid by the winner is equal to the second-highest bid.

This mechanism can be simplified through a simple formulation. Let b_i represent the bid submitted by a participant and v_i represent the true valuation of the item for that participant. The optimal strategy is for each participant to bid their true value ($b_i = v_i$). Bidding below this value reduces the probability of winning without affecting the final price if the auction is won. On the other hand, bidding above the true value may result in winning the auction at a price that exceeds the participant's valuation, leading to a loss.

This property makes the model particularly suitable for MAS. In such systems, multiple autonomous agents interact without centralized control. The auction mechanism simplifies decision making, as agents are not required to estimate competitors' behavior or analyze the broader market, instead, each agent relies solely on its own valuation. This significantly reduces computational complexity while ensuring that resources are allocated efficiently to the agents.

For this paper, it was necessary to implement a repeated sealed-bid second-price auction (Vickrey Auction) with $N = 5$ agents over $T = 26$ rounds. In each round $t \in \{1, \ldots, 26\}$, each agent $i \in \{1, \ldots, 5\}$ receives a private valuation $v_{i,t}$ from a deterministic matrix $V \in \mathbb{R}^{26\times 5}$ (Games A–Z).

Bids are generated via strategy π_i: $b_{i,t} = \pi_i(v_{i,t}, \mathcal{H}_t)$, where $\mathcal{H}_t$ is history.

Winner: $i^* = \arg\max_i b_{i,t}$, price: $p_t = \max_{j\neq i^*} b_{j,t}$.

Payoff: $u_{i^*,t} = v_{i^*,t} - p_t$, and $u_{j,t} = 0$ for $j \neq i^*$.

Agent Strategies:

- **Truthful:** $b_{i,t} = v_{i,t}$;
- **Random:** $b_{i,t} \sim \mathcal{U}(0.85v_{i,t}, 1.15v_{i,t})$;
- **Underbid:** $b_{i,t} = v_{i,t}\theta,\ \theta \sim \mathcal{U}(0.85, 1)$;
- **Overbid:** $b_{i,t} = v_{i,t}\theta,\ \theta \sim \mathcal{U}(1, 1.15)$;
- **Noisy rationality:** $b_{i,t} = v_{i,t} + \epsilon,\ \epsilon \sim \mathcal{N}(0, 0.1v_{i,t})$.

3.4.1. Literature Results Analysis

According to the paper "**A Theoretical and Experimental Analysis of Bidding Behavior in Vickrey-Auction Games**" by "*Werner Güth, Rolf Schmittberger and Bernd Schwarze*", Vickery auction game is best suitable for general simulations and experiments, by citing "*we would like to mention that the Vickrey Auction and probably auction-games in general are extremely suitable for experimental purpose*" [7].

On their paper, the authors analysed and exposed some Vickery Auction experiments results, as exposed on the following table. The original paper values were multiplied by 100 to convert the values from units to hundreds for a slightly better practical representation.

Table 5. Paper Naive Behavior in the Vickrey Auction Game Experiments

Game	v_1	b_1	v_2	b_2	v_3	b_3	v_4	b_4	v_5	b_5
A	135	134	950	750	275	195	500	501	675	660
B	825	600	785	835	100	099	295	251	455	350
C	505	175	655	700	245	242	805	500	255	115
D	445	445	835	900	655	654	645	999	115	100
E	455	407	995	540	655	310	105	104	255	250
F	1000	555	800	779	500	450	400	200	200	199
G	200	180	400	400	600	400	700	550	950	650
H	275	250	455	300	990	920	655	600	245	246
I	105	090	255	210	455	402	1000	443	655	500
J	800	801	650	600	1000	1000	250	200	450	650
K	235	235	435	395	635	450	735	735	935	600
L	795	425	100	150	256	101	545	450	995	995
M	245	170	795	595	1000	980	255	405	500	100
N	900	980	800	360	600	800	400	395	200	050
O	575	150	785	722	995	902	250	250	450	250
P	100	150	250	200	450	400	750	550	963	763
Q	250	200	150	100	550	850	950	888	650	150
R	880	480	420	320	575	400	715	716	185	050
S	935	563	215	400	550	560	750	680	325	324
T	175	100	875	775	530	150	720	720	445	445
U	255	887	650	450	805	550	795	300	100	080
V	995	888	655	355	455	399	255	200	245	200
W	795	850	805	800	105	100	645	300	445	295
X	245	887	255	200	945	398	795	590	500	175
Y	645	280	825	650	100	050	800	675	445	440
Z	500	350	800	790	400	210	1000	500	250	100

Table 6. Simulation Bidding Data

Game	v_1	b_1	v_2	b_2	v_3	b_3	v_4	b_4	v_5	b_5	Winner_ID
A	135	135	950	1012	275	274	500	564	675	575	2
B	825	825	785	713	100	93	295	327	455	540	1
C	505	505	655	717	245	244	805	851	255	294	4
D	445	445	835	723	655	568	645	717	115	103	2
E	455	455	995	993	655	654	105	118	255	241	2
F	1000	1000	800	789	500	426	400	459	200	186	1
G	200	200	400	418	600	511	700	774	950	991	5
H	275	275	455	411	990	893	655	711	245	225	3
I	105	105	255	266	455	424	1000	1076	655	713	4
J	800	800	650	610	1000	877	250	267	450	426	3
K	235	235	435	462	635	549	735	737	935	817	5
L	795	795	100	100	256	239	545	580	995	948	5
M	245	245	795	818	1000	853	255	279	500	532	3
N	900	900	800	692	600	544	400	410	200	204	1
O	575	575	785	856	995	862	250	268	450	450	3
P	100	100	250	247	450	397	750	856	963	871	5
Q	250	250	150	128	550	528	950	973	650	635	4
R	880	880	420	366	575	532	715	726	185	220	1
S	935	935	215	230	550	498	750	815	325	330	1
T	175	175	875	954	530	456	720	722	445	435	2
U	255	255	650	720	805	720	795	853	100	119	4
V	995	995	655	584	455	436	255	290	245	230	1
W	795	795	805	686	105	101	645	654	445	419	1
X	245	245	255	290	945	890	795	859	500	451	3
Y	645	645	825	840	100	91	800	886	445	526	4
Z	500	500	800	797	400	348	1000	1137	250	295	4

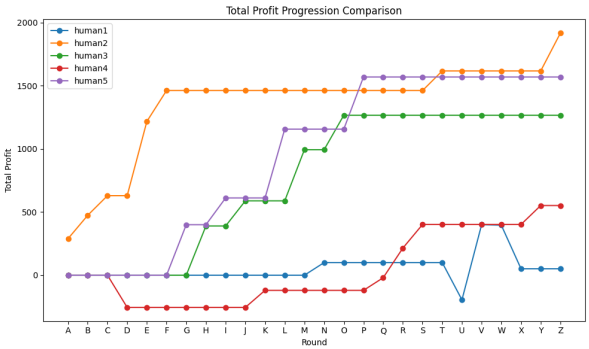


Fig. 5. Paper Game Experiments Player Cumulative Profits

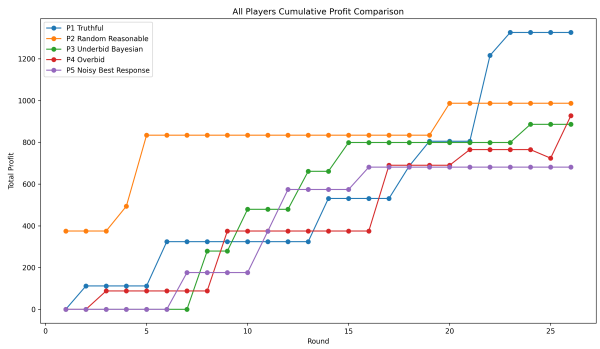


Fig. 6. Simulation Players Cumulative Profit Graph

The simulation results confirm the main theoretical property of the Vickrey Auction, that is *truthful bidding* remains a stable and rational strategy for utility maximization. **The observed outcomes show that winners were generally agents with the highest or near-highest true valuations**, which preserves allocative efficiency even when some participants deviated from truthful bidding.

From a Multi-Agent Systems perspective, these results support the applicability of Vickrey Auctions for decentralized resource allocation, as exposed in book "**Issues in computational Vickrey auctions**" by "*Sandholm Tuomas* [15]. Since agents only require knowledge of their own valuation and not the strategic intentions of others, computational complexity remains low while incentive compatibility is preserved.

3.4.2. Comparison with Literature Review

A direct comparison between the literature results and the simulation highlights the difference between naive human bidding behavior and strategy-driven agent bidding. In the experimental results presented by Güth, Schmittberger and Schwarze [7], participants frequently deviated from the dominant truthful strategy. These deviations indicate that human participants often misunderstand the incentive structure of the Vickrey Auction or are influenced by psychological factors such as risk aversion, competitiveness or attempts to strategically manipulate outcomes. In contrast, the simulated environment produced significantly more structured and rational behavior.

Another important difference lies in reproducibility. Human experiments introduce behavioral noise that is difficult to predict, while the multi-agent simulation provides deterministic valuations and formally defined bidding policies, allowing consistent evaluation of strategic performance. This makes simulation-based analysis more suitable for validating auction mechanisms in computational environments.

4. Conclusion

This research paper explored the relationship between Multiagent Systems (MAS) and Game Theory (GT), showing how these two fields naturally support each other in solving complex decision-making problems. MAS provides a practical framework where multiple autonomous agents can interact, while GT offers the mathematical foundation for analysing strategic behaviour between those agents.

The study reviewed the main theoretical concepts of both fields, including cooperative and non-cooperative games, Nash equilibrium, coordination problems and auction mechanisms. It also presented important real-world applications such as electronic commerce, automated negotiation, bounded rationality and decentralized resource allocation, where MAS and GT are commonly used together.

The two selected study cases, the Minimum Effort Game and the Vickrey Auction, demonstrated how multi-agent simulations can be used to validate theoretical models. In the Minimum Effort Game, agents gradually converged toward the lowest-effort equilibrium, confirming the coordination difficulties described in the literature. In the Vickrey Auction simulation, truthful bidding proved to be the most stable and rational strategy, supporting the theoretical advantages of second-price auctions for resource allocation.

Overall, the results show that MAS are highly effective for modelling and simulating strategic interactions in GT. They allow researchers to study agent behaviour in controlled environments and provide valuable insights into cooperation, competition and equilibrium formation. Future work could extend these simulations by introducing adaptive learning agents, larger-scale environments and more complex real-world scenarios where strategic reasoning plays an important role.

5. Acknowledgement

This work is the result of my own activity, and I confirm I have neither given, nor received unauthorized assistance for this work. I declare that I used generative AI or automated tools in the creation of content or drafting of this document. All the text impacted by generative AI has been explicitly marked as such in this document.

6. Turnitin Score

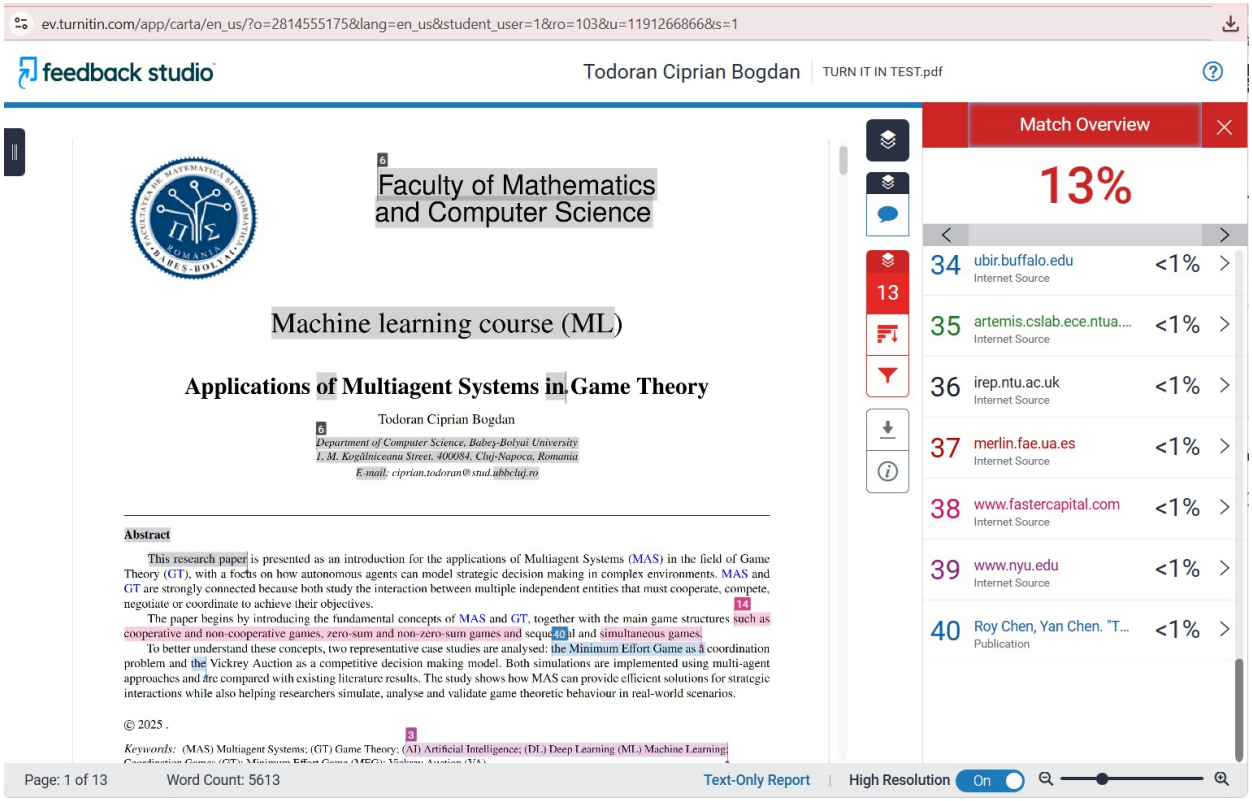


Fig. 7. Turnitin Score of 13 percent similarity

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Acronyms

AI Artificial Intelligence. 2

GT Game Theory. 1–6, 13

MAS Multiagent Systems. 1, 2, 4–7, 11, 13